



LEEDS CITY COUNCIL

Highways and Transportation

Highways Services
St George House
40 Great George Street
Leeds LS1 3DL



Junction: A63 Selby Road / Ninelands Lane

Company: Dynniq

Controller Type: PTC1

Mains Supply: 230V / 50Hz

New / ~~Modification~~

Operating Voltage: ELV

Engineer: Corey Hotchen

Telephone No: 07566 759021

OTU SCN 20150

Cont SCN 20151

IP Address Information:

Controller IP: 10.93.41.65

Gateway: 10.93.41.126

Mask: 255.255.255.192

OTU IP: 10.93.41.66

NOTE TO UTC PERSONNEL This specification should be used for general purposes only.
The Signal Company documentation should be used for controller operation
and timings unless amended on this sheet as RAM data.

RAM UPDATE (If no data then write NONE)	Reasons for change	Initials	Date
Write date current EPROM fitted:-			

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950L A63 Selby Road / Ninelands Lane

General

A Swarco PTC1 8-phase controller from contract 3548 shall provide the method of control in this specification.

The site is to be refurbished with MOVA control added and the pedestrian facilities upgraded to puffin type crossings.

The normal method of control shall be MOVA / VA.

Hurry Call Facility

The controller will be equipped with a hurry call facility from the nearby fire station which shall place a demand for stage 3 dummy phase J. The details of the hurry call timers are as specified in the relevant TR2210 specification forms.

If the hurry call button in the fire station were to fail active then, after a period of 10 minutes, the DFM lamp should be illuminated and stage 3 becomes inactive until the fault has been reset.

Kerbside Detector Operation

The kerbside detectors will operate in 2 ways:

1. To eliminate the requirement for pedestrians to place demands using push buttons the kerbside detectors shall place demands for each respective puffin phase, initially configured with a call delay of 2 seconds and a cancel period of 2 seconds (alterable by handset). This operation shall be configured with a software switch that can be switched by timetable and initially set to active.
2. The general operation of the kerbside and pushbutton demands will be as set out in TOPAS 2500. Pushbutton demands shall be held and cancelled depending on the active state of the kerbside and any push button demand placed when no kerbside presence is detected shall place a latched demand.

NOTE: On DFM fail the kerbside detectors shall not insert demands for their respective puffin phases. Latched push button demands will be required. Software switches to be available to disable kerbside demands.

MOVA Configuration

MOVA UTC Interface:

Control Word: F1 F2 F3 F4 F5 F6 TO
G1 G2 G3 G4 G5 G6 CRB MO

Wait Lamp Mimic: Whilst pedestrian phase wait lamp is lit, input to MOVA:

Ped F = Det 17

Ped G = Det 18

Ped H = Det 19

CRB toggle: CRB configured to toggle every 120 seconds if MOVA is not in control.

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Pedestrian Forward Calling Issue 3 update - March 2024

Forward calls to enable pedestrian progression shall be demanded by push button pole 7 for northbound progression and push button pole 12 for southbound progression. Each forward calling demand shall be switchable and have a dedicated output to MOVA that pulses the following outputs pole 7 MOVA det. 21 and pole 12 MOVA det. 22..

The three possible demand conditions shall provide the following: -

Demand at pole 7 ONLY - Southbound **G --> F - (17.25m walk@1.2m/s = 15 secs)** **A demand for ped phase G placed at pole 7 shall also place a demand for stage 5 and dummy phase L, stage 5,**
To ensure progression the minimum duration of stage 5 shall be fixed by **phase F min green (appears only on demand for min green period only, initially set to 5 secs).**

Demand at pole 12 ONLY - Northbound - **F --> G (15.3m walk@1.2m/s = 13 secs)** **As well as demanding ped phase F, stage 4, a demand shall be placed for dummy phase K.**
To ensure progression the minimum duration of stage 4 shall be fixed by **phase K min green (appears only on demand for min green period only, initially set to 5 secs).**

Demand at pole 7 & pole 12 - Northbound and Southbound - Same as Northbound only, as this caters for both pedestrian movements.

MOVA Loop Positions

Detectors shall be wired into packs as shown below, to prevent chattering.

To assist in validation all detector inputs to be clearly labelled on detector packs in controller.

Approach	Loop	Dist. From stop line (m)	Longitudinal Dimension (m)	Shape	Clearance Distance		Detector. Pack No.
					N/S	O/S	
A63 Selby Road Eastbound	AIN1	90	1.7	Diamond	0.4	0.8	1
	AX2	40	1.7	Diamond	0.4	0.8	1
	IX3	25	1.7	Diamond	0.4	0.8	1
	ISL4	1	2	Rectangular	0.4	0.4	1
	ASL5	1	2	Rectangular	0.4	0.4	2
A63 Selby Road Westbound	BIN6	88	1.7	Diamond	0.4	0.8	2
	BX7	40	1.7	Diamond	0.4	0.8	2
	BX8	25	1.7	Diamond	0.4	0.8	2
	BSL9	1	2	Rectangular	0.4	0.4	3
	BSL10	1	2	Rectangular	0.4	0.4	3
	COUT11	N/A	N/A	Rectangular	0.4	0.4	3
Ninelands Lane Southbound	EIN12	78	1.7	Diamond	0.4	0.8	3
	EX13	42	1.7	Diamond	0.4	0.8	4
	EX14	20	1.7	Diamond	0.4	0.8	4
	ESL15	1	2	Rectangular	0.4	0.4	4
	ESL16	1	2	Rectangular	0.4	0.4	4

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Flow / Occupancy Loops

Two outputs shall be configured for loops AIN1 (SCN 44318), BIN6 (SCN 44319), EIN12 (SCN 44320), EX14 (SCN 44321) and shall generate replied on the corresponding bits as allocated in the bit pattern.

Flow loops shall change the state of the respective reply bit after every 4 vehicles that pass over the loop.

Occupancy loops shall be configured to simply reply the raw input.

Corridor Control Strategy

The following control bits are processed by UTC and sent based on different traffic conditions on the corridor. The purpose of these is to hold the crossing at green for vehicles to flush traffic through when required.

If control bit SF1 is received, pulse MOVA Det 41 for 2 seconds. (SCN 44310)

If control bit SF2 is received, pulse MOVA Det 42 for 2 seconds. (SCN 44311)

If control bit SF3 is received, pulse MOVA Det 43 for 2 seconds. (SCN 44312)

If control bit SF4 is received, pulse MOVA Det 44 for 2 seconds. (SCN 44313)

Dynamic Dataset Changes

Based on the flow/occupancy/queueing conditions of the corridor it is desirable to change MOVA Datasets. The following logic and MOVA detectors facilitate the changes.

If control bit DS1 is being received from UTC, output to MOVA Det. 37 (SCN 44314)

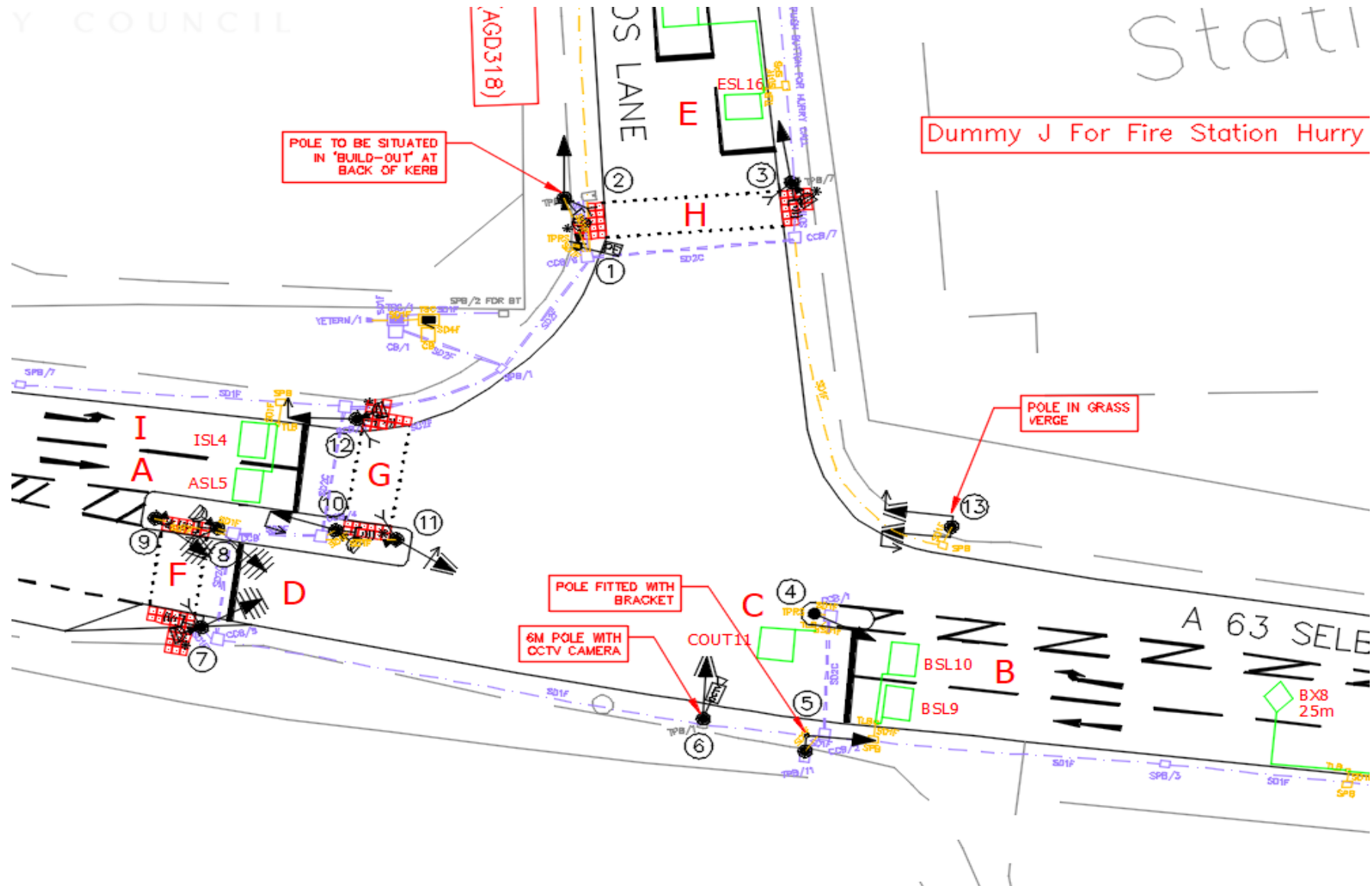
If control bit DS2 is being received from UTC, output to MOVA Det. 38 (SCN 44315)

If control bit DS3 is being received from UTC, output to MOVA Det. 39 (SCN 44316)

If control bit DS4 is being received from UTC, output to MOVA Det. 40 (SCN 44317)

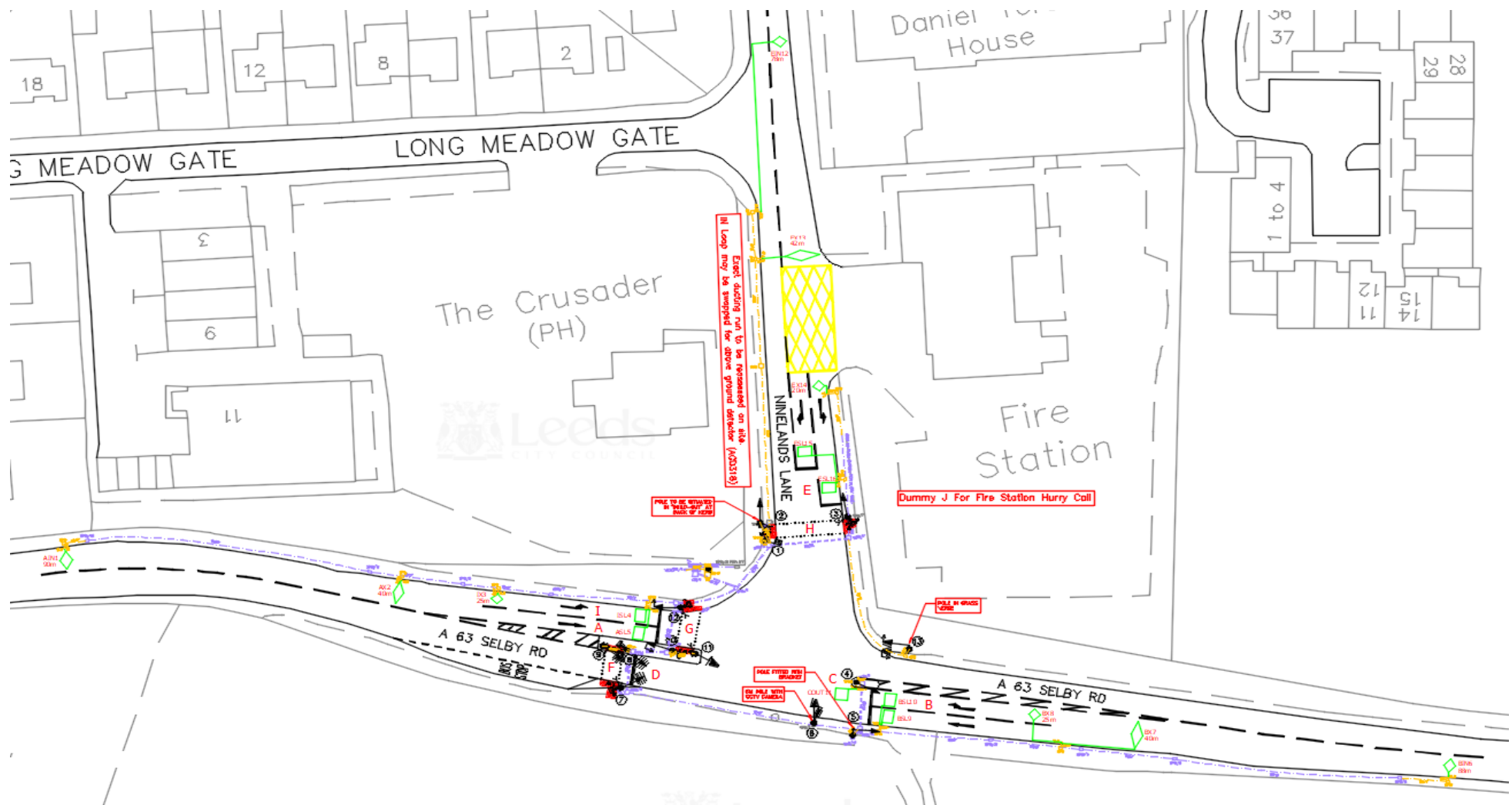
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Site Layout



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Site Layout



NOT TO SCALE

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USE OF PHASES		PHASE	APPEAR	TERMINATION		PED	PERIOD	MIN	MAXIMUMS					
									(Day)	(AM)	(PM)	(Night)	(Weekend)	
				TYPE	ASSOC				SET	SET	SET	SET	SET	SET
LOCATION		NOTE 3	NOTE 1	NOTE 2	PHASE	BLKT	V	GREEN	A	B	C	D	E	F
A	A63 Selby Road, Eastbound	T	0	0				7	30	40	40	30		
B	A63 Selby Road, Westbound	T	0	0				7	30	40	40	30		
C	A63 Selby Road, Westbound Right	I	0	2	B			4	10	10	12	10		
D	A63 Selby Road, Westbound 2nd Stopline	T	0	0				7	-	-	-	-		
E	Ninelands Lane, Southbound	T	0	0				7	15	20	20	15		
F	Ped Crossing, Westbound	P	0	0				5						
G	Ped Crossing, Eastbound	P	0	0				5						
H	Ped Crossing, Ninelands Lane	P	0	0				5						
I	A63 Selby Road, Eastbound Left	T	0	0				7	10	10	12	10		
J	Dummy, Stage 4 HURRY CALL	D	1	0				7						
K	Dummy, Stage 4 NTH PED PROGRESSION	D	0	0				2						
L	Dummy, Stage 5 STH PED PROGRESSION	D	0	0				2						
M	Dummy, Stage 6 QAR	D	0	0				2						

Note 1:- Condition of Appearance

0 - Phase always appears when stage runs

1 - Phase only appears if demand exists at start of inter-stage

2 - Phase appears, if demanded, at any time up until the end of the stage in which the phase runs

3 - Phase appears, if demanded at any time during the stage up until the window time expires

Note 2:- Condition of Termination

0 - Terminates at end of stage

1 - Terminates when associated phase gains right of way

2 - Terminates when associated phase loses right of way

3 - Terminates at end of minimum green

4 - Terminates at end of maximum green

5 - Terminates subject to special conditioning

Note 3:- Phase Types

T - Traffic F - Filter Arrow

PD- Pedestrian I - Indicative Arrow

PU- Puffin D - Dummy

TN - Toucan Nearside L - L.R.T

TF - Toucan Farside S - Switched Sign

C - Cycle

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[illegible][illegible][illegible]

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[illegible]

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[illegible]

C = CONFLICTING & OPPOSING
O = OPPOSING ONLY
. = NO CONFLICT

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ADDITIONAL PHASE DELAYS														
No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs	No	Delay Phase	Change From	To Stage	By (X) Secs
1	D	1	3	6	29					57				
2	D	2	3	6	30					58				
3	D	3	1	+1	31					59				
4	F	3	4	5	32					60				
5	D	3	5	+5	33					61				
6	B	4	1	+2	34					62				
7	D	4	3	6	35					63				
8					36					64				
9					37					65				
10					38					66				
11					39					67				
12					40					68				
13					41					69				
14					42					70				
15					43					71				
16					44					72				
17					45					73				
18					46					74				
19					47					75				
20					48					76				
21					49					77				
22					50					78				
23					51					79				
24					52					80				
25					53					81				
26					54					82				
27					55					83				
28					56					84				

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(Insert intergreen diagrams here)

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(Insert intergreen diagrams here)

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(Insert intergreen diagrams here)

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PROHIBITED / ALTERNATIVE STAGE MOVEMENTS UP TO 16 STAGES

[illegible]

TICK AS REQUIRED		
MODE	These Restrictions are for	No Restrictions
Urban Traffic Control		
Cableless Linking		
MOVA		
Hurry Call	✓	
Manual	✓	
Priority		
Emergency		
VA	✓	

NOTE:

✓	= Permitted
P	= Prohibited
X	= Ignore
N'	= ALTERNATIVE MOVES

where 'N' is number of stage to move to

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS

ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
1	AIN1	V		A				4.0			1		
2	AX2	V	A	A				4.0			1		
3	IX3	V	I	I				4.0			1		
4	ISL4	V	I	I		3		0.5			1		
5	ASL5	V	A	A		3		0.5			1		
6	BIN6	V		B				4.0			1		
7	BX7	V	B	B				4.0			1		
8	BX8	V	B	B				4.0			1		
9	BSL9	V	B	B		3		0.5			1		
10	BSL10	V	B	B		3		0.5			1		
11	COUT 11	V	C	C	✓	4	2	2			1		
12	EN12	V		E				4.0			1		
13	EX13	V	E	E				4.0			1		
14	EX14	V	E	E				4.0			1		
15	ESL15	V	E	E		3		0.5			1		
16	ESL16	V	E	E		3		0.5			1		
17	PBU F - Pole 7	PB	F		✓						3	Y	
18	Kerbside - Pole 7	K							17		1	A	
19	On-crossing - Pole 7	O						0.5			1	A	
20	PBU F - Pole 8	PB	F		✓						3	Y	

✓	Tick appropriate boxes	
Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-			Error States
V Vehicle	C Clearance	SU Speed Upstream	A Fail Active
PB Pushbutton	Q Queue	SD Speed Downstream	I Fail Inactive
K Kerbside	U1 Uni Directional 1		Y Fail in input state
O On-Crossing	U2 Uni Directional 2		
Paired Detector:-			
Specify ID of paired Uni-directional or kerbside/pushbuttons			

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS

ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
21	Kerbside - Pole 8	K							20/23		1	A	
22	On-crossing - Pole 8	O						0.5			1	A	
23	PBU F - Pole 9	PB	F		✓						3	Y	
24	PBU G - Pole 10	PB	G		✓						3	Y	
25	Kerbside - Pole 10	K							24/27		1	A	
26	On-crossing - Pole 10	O						0.5			1	A	
27	PBU G - Pole 11	PB	G		✓						3	Y	
28	PBU G - Pole 12	PB	G		✓						3	Y	
29	Kerbside - Pole 12	K							28		1	A	
30	On-crossing - Pole 12	O						0.5			1	A	
31	PBU H - Pole 1	PB	H		✓						3	Y	
32	Kerbside - Pole 1	K							31/34		1	A	
33	On-crossing - Pole 1	O						0.5			1	A	
34	PBU H - Pole 2	PB	H		✓						3	Y	
35	PBU H - Pole 3	PB	H		✓						3	Y	
36	Kerbside - Pole 3	K							35		1	A	
37	On-crossing - Pole 3	O						0.5			1	A	
38	Tactile Fault - Pole 1										4	A	
39	Tactile Fault - Pole 3										4	A	
40	Tactile Fault - Pole 7										4	A	

✓	Tick appropriate boxes	
Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-			Error States
V Vehicle	C Clearance	SU Speed Upstream	A Fail Active
PB Pushbutton	Q Queue	SD Speed Downstream	I Fail Inactive
K Kerbside	U1 Uni Directional 1		Y Fail in input state
O On-Crossing	U2 Uni Directional 2		
Paired Detector:-			
Specify ID of paired Uni-directional or kerbside/pushbuttons			

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DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS

ID	Location / Title	Type	Demand	Extend	Non - Latching	Call Delay	Cancel Delay	Extension	Paired Detector	Kerbside Extension	DFM Group	Error State	Special Instructions
41	Tactile Fault - Pole 9										4	A	
42	Tactile Fault - Pole 11										4	A	
43	Tactile Fault - Pole 12										4	A	
44													
45													
46													
47													
48													
49													
50													
51													
52													
53													
54													
55													
56													
57													
58													
59													
60													

☒	Tick appropriate boxes	
Group	Detector Fault Monitoring Times (DFM)	
	Fail Active after (mins)	Fail Inactive after (hours)
1	30	18
2	10	24
3	10	0
4	10	120

Detector Types:-			Error States
V Vehicle	C Clearance	SU Speed Upstream	A Fail Active
PB Pushbutton	Q Queue	SD Speed Downstream	I Fail Inactive
K Kerbside	U1 Uni Directional 1		Y Fail in input state
O On-Crossing	U2 Uni Directional 2		
Paired Detector:-			
Specify ID of paired Uni-directional or kerbside/pushbuttons			

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MOVA DETECTORS								
No.	Name / Description	Action	No.	Name / Description	Action	No.	Name / Description	Action
1	AIN1		25	DS1	MOVA Dataset 1	49		
2	AX2		26	DS2	MOVA Dataset 2	50		
3	IX3		27	DS3	MOVA Dataset 3	51		
4	ISL4		28	DS4	MOVA Dataset 4	52		
5	ASL5		29	SF1		53		
6	BIN6		30	SF2		54		
7	BX7		31	SF3		55		
8	BX8		32	SF4		56		
9	BSL9		33			57		
10	BSL10		34			58		
11	COU11		35			59		
12	EN12		36			60		
13	EX13		37			61		
14	EX14		38			62		
15	ESL15		39			63		
16	ESL16		40			64		
17	PBU F		41			65		
18	PBU G		42			66		
19	PBU H		43			67		
20	All Red Revert K	QAR	44			68		
21	Pole 7 Demand	Nrthbnd ped progress	45			69		
22	Pole 12 Demand	Sthbnd ped progress	46			70		
23			47			71		
24			48			72		

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ADDITIONAL SWITCHES & LED'S (ON MANUAL PANEL)		
POSITION	USE	MOMENTARY *
SWITCH 1		
LED 1	MOVA Enabled (Flashing) UTC Mode (Solid) Stream 1	
SWITCH 2	DEMAND HURRY CALL	✓
LED 2	HURRY CALL ACTIVE	
SWITCH 3		
LED 3		
HURRY CALL AUX 4		
HIGHER PRIORITY AUX 5		

HURRY CALL(S)					
HC(0) ALWAYS HAS PRIORITY OVER HC(1)					
HURRY CALL No	CALL STAGE	CALL DELAY	CALL HOLD	CALL PREVENT	NOTES
1	3	8	32	120	
CALL LIMIT	LOW	0	0	60	
	HIGH	60	60	240	

OPTION FILTER GREEN ARROW IN MANUAL			
TO APPEAR	NOT TO APPEAR	DEMAND DEPENDANT	INSERT TICK

PART TIME MODE	
SWITCH OFF IN STAGE(S)	

*

NORMAL ACTION IS A TOGGLE ACTION
(PRESS ONCE FOR ACTIVE, PRESS AGAIN TO MAKE INACTIVE)

PUT TICK IF MOMENTARY ACTION IS REQUIRED
(ACTIVE ONLY WHILE BUTTON IS PRESSED)

IF SWITCH IS MOMENTARY OR NOT USED THEN THE ASSOCIATED LED IS AVAILABLE FOR OTHER PURPOSES

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EXTEND ALL RED FACILITY - (BY DETECTOR) CONSIDERED AS PART OF INTERGREENS

Stage to Stage transition for which ALL RED extension required					
UNIT No	From Stage	To Stage	Ext. Time	All Red Max	Associated Detector
1	1	3, 4, 5	2	6	COUT11
2					
3					
4					
5					
6					
7					

EXTEND ALL RED TIMINGS				
STREAMS	0	1	2	3
EXTENSION TIME (SECS)*				
ALL RED MAXIMUM TIME (SECS)*				

* Measured from point at which first phase in new stage would normally start its red-green transition period.

RED EXTENSION	ENTER CHOICE
MANUAL	
CLF	
UTC	
HC	
PRIORITY	
EMERGENCY	
VA	
FT	

For Choice, enter as follows:-

O = Operative

M = Extend to Maximum

N = No extensions

(Not approved for UK)

If table is left blank then all modes will be operative.

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MODE PRIORITY

ASSIGN NUMBER 1 = TOP PRIORITY
2 = SECONDY PRIORITY etc.
X = NOT USED

_____ PART TIME
_____ EMERGENCY VEHICLE
1 HURRY CALL(S)
2 URBAN TRAFFIC CONTROL
3 SELECTED MANUAL CONTROL (1)
4 SELECTED FIXED TIME OR VA (2)
6 CABLELESS LINKING
5 MOVA
7 EITHER VEHICLE ACTUATED
_____ OR FIXED TIME

PERMANENTLY DISABLED MODES

MODES TO BE DISABLED	STREAM			
	0	1	2	3
PART TIME (3)				
EMERGENCY VEHICLE				
HURRY CALL(S)				
SELECTED MANUAL CONTROL (3)				
URBAN TRAFFIC CONTROL				
SELECTED FIXED TIME OR VA				
CABLELESS LINKING				
BUS PRIORITY				
VEHICLE ACTUATED				
FIXED TIME				
✓ TICK IF REQUIRED TO BE DISABLED.				

MANUAL FACILITY ENABLED

ALWAYS	✓
WHEN HANDSET PLUGGED IN	
WHEN "MND" COMMAND ENTERED	

FACILITY SWITCH POSITION	NOT OPERATIVE INSERT (X)
VA	
FIXED TIME	
CLF	

(3) THESE MODES MAY ONLY BE DELETED ON ALL MODES SIMULTANEOUSLY

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FIXED CYCLE TIME MODE STAGE SEQUENCE								
Phase	A	B	C	D	E	F	G	H
Stage	1st	2nd	3rd	4th	5th	6th	7th	8th
Time								

OR

ALTERNATIVELY FIXED CYCLE RUNNING TO CURRENT TIMETABLED MAXIMA																
Phase	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
Demand Dependent			4			4	4	4								
Phase	Q	R	S	T	U	V	W	X	Y	Z						
Demand Dependent																

MANUAL SELECTION		
Button No.	Stage(s) called	Street Names - If Required
0	7	MANUAL ALL RED
1	1	A63 SELBY RD
2	2	A63 SELBY RD, WESTBOUND ECO
3	3	NINELANDS LN
4	4	PEDESTRIAN STAGE - ALL ROUND

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URBAN TRAFFIC CONTROL																	
USE OF BITS								OTU MANUFACTURER & TYPE:-									
Bit No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	
Control	SO	F1	F2	F3	F4	F5	F6	D2	D3	D4	D5	GO1	SF1	SF2	SF3	SF4	
Reply	MC	G1	G2	G3	G4	G5	G6	DR2	DR3	DR4	DR5	MO1	DF	LF1	LF2	FL1	
Secret Sign																	
Secret Sign with security																	Related Phase/Stage
Bit No	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
Control	DS1	DS2	DS3	DS4													
Reply	OC1	FL2	OC2	FL3	OC3	FL4	OC4										
Secret Sign																	
Secret Sign with security																	Related Phase/Stage
Notes:- Control: SF Special facility used for Stage Holds, DS used for Dynamic Dataset changes Replies: SC is Scoot bit (VS), FL is Vehicle Count bit (VC), OC is Vehicle Occupancy Bit (VO), Q is Queue loop, MO to reply when Mova is Offline.																	
0141 CONTROL AND REPLY BIT NOTATIONS																	
CONTROL BITS Force (F1,F2,etc) Stage Demand (D1,D2,etc) Common Demand (DX) SW FAC (SF1,SF2) CLF Group Sync (SG) RTC Sync (TS or TS1) Fall Back Mode Select (FM) Lamps On (LO) Local Link Inhibit (LL) Transmission Confirm (TC) Solar Switch Over-ride (SO) Pedestrian Demand (PX) Hold Vehicle Stage (PV) Solar Switch Over-ride (SO)								REPLY BITS Stage Confirm (G1,G2,etc) Stage Dem Confirm (DR1,DR2,etc) SW FAC Confirm (SC1,SC2,etc) CLF Group Sync Confirm (CS) CLF Group 0 Confirm (GR1) RTC Sync Confirm (CC or CC1) Fall Back Confirm (FC) Lamps Off Confirm (LE) Hurry Call Confirm (HC) Manual Control Mode (MC) Controller Fault (CF) Detector Fault (DF) Remote Reconnect (RR) One or more Signal Lamps Failed (LF) Pedestrian Inhibit by Lamp Monitor (PI) Pedestrian Reply (GX)								UTC CONTROL REQUIRED WHEN:- ☒ Force Bits Present ☐ Force Bits and TC/TO Present ☐ TC/TO Present B.T. Termination requirements are:- B.T. Line Number:- 	

(delete one)

[illegible][illegible]

DR1	DR2	DR3	DR4	DR5	DR6	DR7	DR8	DR9	DR10	DR11	DR12
	2	3	4	5							

[illegible]

1. Hash (#) indicates demand dependent force

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UTC ONLY - MANUAL MODE, NO LAMP SUPPLY, LAMP FAILURES etc., CONFIRM						
TICK AS APPROPRIATE:-	G1/G2	G6/G7	MC	DF	LF1	LF2
MANUAL MODE SELECTED AND OPERATIVE	✓	✓	✓			
NO LAMP POWER and/or SIGNALS OFF	✓	✓				
DETECTOR FAULT				✓		
SELECT SWITCH NOT IN NORMAL POSITION			✓			
ANY LAMP FAILURE					✓	
2nd RED LAMP FAILURE ON ANY PHASE					✓	✓

Leeds UTC Lamp Monitoring Specification:

On any lamp failure the LF1 reply bit shall be returned.

On the 1st red lamp failure to any phase conflicting with a pedestrian phase, the LF1 reply bit shall be returned and the detector fault monitor lamp shall flash.

On any second red lamp failure to any phase LF2 shall be returned.

On any second red lamp failure to any phase conflicting with a pedestrian phase, the LF2 reply bit shall be sent, the detector fault monitor lamp shall continue to flash, the conflicting pedestrian phase wait indicators shall be illuminated and the conflicting pedestrian phase demands shall be inhibited. All other non-conflicting pedestrian phases shall continue to be serviced.

On any lamp failure there shall be no intergreen extensions.

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Dyinniq (Peek) Timetable Data

ID.	Function	Arg1	Arg2	Week Days & Special	Start	End	Notes
1	Timing Set	1	0	MTWTF__	07:00:00		Maxset A
2	Timing Set	2	0	MTWTF__	09:30:00		Maxset B
3	Timing Set	3	0	MTWTF__	18:00:00		Maxset D
4	Timing Set	4	0	MTWT__	15:30:00		Maxset C
5	Timing Set	4	0	____F__	15:00:00		Maxset C
6	Timing Set	6	0	____S	10:00:00	18:00:00	Maxset E
7	Timing Set	7	0	____S	10:00:00	16:00:00	Maxset F
8							
9							
10							

Option	Function	Arguments - Arg1 , Arg2
1	Start up Control Plan	Arg1: Control plan number, always 1
2	Select a phase Timing Set	Arg1: Timing Set Number
3	Part-time signal (auto, flash, off)	Arg1: Stream Number (from 1), 0 = all streams Arg2: 0 = Auto, 1 - flash (not allowed in UK), 2 = Off
4	Select a CLF plan	Arg1: CLF plan number (from 1)
5	Isolate a standalone stream	Arg1: 0 = No, 1 = Yes
6	Enable / Disable Audible signals	Arg1: 0 = Off, 1 = On
7 - 30	Defined User Function plans	User defined arguments
31	Dimming Plan	User defined arguments

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Plan : Cycletime : Offset :					Plan : Cycletime : Offset :					Plan : Cycletime : Offset :					Plan : Cycletime : Offset :				
Act	Influence	Stage	Stream	Time	Act	Influence	Stage	Stream	Time	Act	Influence	Stage	Stream	Time	Act	Influence	Stage	Stream	Time
1					1					1					1				
2					2					2					2				
3					3					3					3				
4					4					4					4				
5					5					5					5				
6					6					6					6				
7					7					7					7				
8					8					8					8				
9					9					9					9				
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21					21					21					21				
22					22					22					22				
23					23					23					23				
24					24					24					24				

SPECIAL CONDITIONING - GENERAL REQUIREMENTS	
Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLELESS LINKING	
1 - 2	$(\bar{F}1 \cdot F2 + F1 \cdot F2 \cdot GO \cdot \bar{EXT} A \cdot \bar{EXT} B \cdot \bar{EXT} I) \cdot (DEM C + D2)$
1 - 3	$(\bar{F}1 \cdot F3 + F1 \cdot F3 \cdot GO \cdot \bar{EXT} A \cdot \bar{EXT} B \cdot \bar{EXT} I) \cdot (DEM E + DEM G + DEM J + D3)$
1 - 4	$(\bar{F}1 \cdot F4 + F1 \cdot F4 \cdot GO \cdot \bar{EXT} A \cdot \bar{EXT} B \cdot \bar{EXT} I) \cdot (DEM F + DEM K + D4)$
1 - 5	$(\bar{F}1 \cdot F5 + F1 \cdot F5 \cdot GO \cdot \bar{EXT} A \cdot \bar{EXT} B \cdot \bar{EXT} I) \cdot (DEM H + DEM L + D5)$
2 - 3	$(\bar{F}2 \cdot F3 + F2 \cdot F3 \cdot GO \cdot \bar{EXT} B \cdot \bar{EXT} C) \cdot (DEM E + DEM G + DEM J + D3)$
2 - 4	$(\bar{F}2 \cdot F4 + F2 \cdot F4 \cdot GO \cdot \bar{EXT} B \cdot \bar{EXT} C) \cdot (DEM F + DEM K + D4)$
2 - 5	$(\bar{F}2 \cdot F5 + F2 \cdot F5 \cdot GO \cdot \bar{EXT} B \cdot \bar{EXT} C) \cdot (DEM H + DEM L + D5)$
3 - 4	$(\bar{F}3 \cdot F4 + F3 \cdot F4 \cdot GO \cdot \bar{EXT} E) \cdot (DEM F + DEM K + D4)$
3 - 5	$(\bar{F}3 \cdot F5 + F3 \cdot F5 \cdot GO \cdot \bar{EXT} E) \cdot (DEM H + DEM L + D5)$
3 - 1	$(\bar{F}3 \cdot F1 + F3 \cdot F1 \cdot GO \cdot \bar{EXT} E) \cdot (DEM I)$
3 - 2	$(\bar{F}3 \cdot F2 + F3 \cdot F2 \cdot GO \cdot \bar{EXT} E) \cdot (DEM C + D2)$
4 - 1	$(\bar{F}4 \cdot F1 + F4 \cdot F1) \cdot (DEM I)$
4 - 2	$(\bar{F}4 \cdot F2 + F4 \cdot F2) \cdot (DEM C + D2)$
4 - 3	$(\bar{F}4 \cdot F3 + F4 \cdot F3) \cdot (DEM E + DEM G + DEM J + D3)$
4 - 5	$(\bar{F}4 \cdot F5 + F4 \cdot F5) \cdot (DEM E + DEM G + DEM L + D5)$
5 - 1	$(\bar{F}5 \cdot F1 + F5 \cdot F1 \cdot GO \cdot \bar{EXT} A) \cdot (DEM I)$
5 - 2	$(\bar{F}5 \cdot F2 + F5 \cdot F2 \cdot GO \cdot \bar{EXT} A) \cdot (DEM C + D2)$
5 - 3	$(\bar{F}5 \cdot F3 + F5 \cdot F3 \cdot GO \cdot \bar{EXT} A) \cdot (DEM E + DEM G + DEM J + D3)$
5 - 4	$(\bar{F}5 \cdot F4 + F5 \cdot F4 \cdot GO \cdot \bar{EXT} A) \cdot (DEM F + DEM K + D4)$

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1 D2 - COMPUTER DEMAND BIT FOR STAGE 2 dem C - DEMAND FOR PHASE C

EXT C - LACK OF EXTENSIONS FOR PHASE C max C - MAX TIMER FOR PHASE C EXPIRED “.” - BOOLEAN “AND” “+” - BOOLEAN “OR”

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS				
Mode during which conditions apply :- VEHICLE ACTUATED / FIXED TIME				
1 - 2	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{I} + \text{MAX } I) \cdot \text{DEM } C$			
1 - 3	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{I} + \text{MAX } I) \cdot (\text{DEM } E + \text{DEM } G + \text{DEM } J)$			
1 - 4	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{I} + \text{MAX } I) \cdot (\text{DEM } F + \text{DEM } K)$			
1 - 5	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{I} + \text{MAX } I) \cdot (\text{DEM } H + \text{DEM } L)$			
2 - 3	$(\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{C} + \text{MAX } C) \cdot (\text{DEM } E + \text{DEM } G + \text{DEM } J)$			
2 - 4	$(\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{C} + \text{MAX } C) \cdot (\text{DEM } F + \text{DEM } K)$			
2 - 5	$(\overline{\text{EXT}} \bar{B} + \text{MAX } B) \cdot (\overline{\text{EXT}} \bar{C} + \text{MAX } C) \cdot (\text{DEM } H + \text{DEM } L)$			
3 - 4	$(\overline{\text{EXT}} \bar{E} + \text{MAX } E) \cdot (\text{DEM } F + \text{DEM } K)$			
3 - 5	$(\overline{\text{EXT}} \bar{E} + \text{MAX } E) \cdot (\text{DEM } H + \text{DEM } L)$			
3 - 1	$(\overline{\text{EXT}} \bar{E} + \text{MAX } E) \cdot \text{DEM } I$			
3 - 2	$(\overline{\text{EXT}} \bar{E} + \text{MAX } E) \cdot \text{DEM } C$			
4 - 1	$(\text{MIN } F + \text{MIN } G + \text{MIN } H) \cdot \text{DEM } I$			
4 - 2	$(\text{MIN } F + \text{MIN } G + \text{MIN } H) \cdot \text{DEM } C$			
4 - 3	$(\text{MIN } F + \text{MIN } G + \text{MIN } H) \cdot (\text{DEM } E + \text{DEM } G + \text{DEM } J)$			
4 - 5	$(\text{MIN } F + \text{MIN } G + \text{MIN } H) \cdot (\text{DEM } H + \text{DEM } L)$			
5 - 1	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot \text{DEM } I$			
5 - 2	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot \text{DEM } C$			
5 - 3	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\text{DEM } E + \text{DEM } G + \text{DEM } J)$			
5 - 4	$(\overline{\text{EXT}} \bar{A} + \text{MAX } A) \cdot (\text{DEM } F + \text{DEM } K)$			